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Wardrobe for a stowage space of a passenger car

Abstract

A wardrobe of a stowage space of a passenger car includes a stowage space lid where a hanger is disposed on an inside of the stowage space lid and where an item of clothing is hangable on the hanger in a hanging position when the stowage space lid is open. The wardrobe further includes a storage device where the item of clothing is displaceable by the storage device, when the stowage space lid is closed, from the hanging position to a storage position on an underside of the inside of the stowage space lid and/or below an upper side of the stowage space.

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Background/Summary

BACKGROUND AND SUMMARY OF THE INVENTION

(1) The invention relates to a wardrobe for a stowage space of a passenger car.

(2) Storing items of clothing in the stowage space or boot of a passenger car is often problematic,

since, for example, there may be other objects, dirt, water or the like in the stowage space or on the floor thereof. This often results in soiling or creasing of the items of clothing. In addition, it is sometimes not possible to fully utilize the capacity of the stowage space when storing items of clothing.

(3) A storage box for a cargo space of vehicles is known from EP 2 574 505 A1, in which a roller blind cover for covering the cargo space can be brought into a position in which it forms a trough-like receptacle on the upper side, in which receptacle items of clothing can be stored. However, such an arrangement only allows the respective items of clothing to be placed above the stowage space and thus also more or less outside the stowage space. Moreover, such a set-up is not suitable for notchback saloons.

(4) The object of the present invention is therefore to create a wardrobe for a stowage space of a passenger car by means of which respective items of clothing can be stored particularly favourably within the stowage space.

(5) The wardrobe according to the invention for a stowage space of a passenger car comprises a stowage space lid which makes the stowage space accessible or closes it and on the inside of which is arranged a hanging means on which at least one item of clothing can be hung in a hanging position when the stowage space lid is open. Furthermore, the wardrobe comprises a storage device by means of which the at least one item of clothing can be displaced, when the stowage space lid is closed, from the hanging position to a storage position at least substantially on the underside of the inside of the stowage space lid and/or of an upper side of the stowage space.

(6) It is accordingly provided according to the invention that respective item of clothing can initially be arranged in a hanging position on the corresponding hanging means when the stowage space lid is open, after which, when the stowage space lid is closed, it can be displaced by means of the storage device from the hanging position into the storage position, in which the item of clothing is arranged in an approximately horizontal position, for example, in the area in proximity to the underside of the inside of the stowage space lid and/or of an upper side of the stowage space. Therefore, when the stowage space lid is closed, the at least one item of clothing is particularly expediently moved into an upper area of the stowage space in which no objects are located and/or no dirty water or the like collects. As a result, the respective items of clothing can be stored carefully and in a particularly expedient manner without them becoming soiled or creased in particular.

(7) In a further embodiment of the invention, it has been shown to be advantageous if the storage device has at least one flexible storage element, the length of which relative to the stowage space lid can be changed with its one end on the inside of the stowage space lid and with its other end by means of a retraction device. By changing the relative length of the flexible storage element, the at least one item of clothing can be displaced from the hanging position to the storage position and vice versa during the closing or opening of the stowage space lid. Flexible in this context means that the respective storage element can be reversibly bent and rolled up without any plastic deformation occurring as result of the bending or rolling up or the like. In particular, textiles are conceivable here, for example textile straps or textile webs, or else strips, webs or the like made of a plastic, a reinforced plastic film or the like.

(8) In a further embodiment of the invention, respective storage straps or the like are provided as flexible storage elements, the length of which relative to the stowage space lid can be changed with their one end on the inside of the stowage space lid and with their other end by means of a respective retraction device. By way of example, two storage straps running parallel to each other can be provided, by means of which the respective items of clothing can be displaced between the hanging position and the storage position with the stowage space lid is closed or opened. Such storage straps can be accommodated in the stowage space in a particularly space-saving manner and, moreover, can be retracted or released particularly expediently by means of respective retraction devices.

(9) A further advantageous embodiment provides that on the side of the at least one item of clothing opposite the at least one flexible storage element, at least one further flexible storage element or respective storage straps is/are provided, the length of which relative to the stowage space lid can be changed with its/their one end on the inside of the stowage space lid and with its/their other end by means of a respective retraction device, wherein the item of clothing is arranged in its storage position between the flexible storage elements or storage straps on both sides. The storage elements or storage straps arranged on both sides thus enable a particularly reliable displacement of the items of clothing between the hanging position and the storage position.

(10) Furthermore, it has been shown to be advantageous if respective transverse storage elements extend between the storage straps. For example, these transverse storage elements can extend between storage straps that run essentially parallel to each other. The transverse storage elements enable a particularly reliable storage of the items of clothing, in particular in their storage position. The transverse storage elements can also be formed, for example, as flexible straps, strips or the like made of a textile or plastic. This results in an overall extremely space-saving and easily stowable arrangement of the storage straps or transverse storage elements. However, it would also be conceivable to use rigid elements, for example made of wood, plastic or metal.

(11) A further advantageous embodiment provides that the storage device has at least one deflector element for the associated flexible storage element or respective storage straps. Thus, the respective item of clothing can be displaced particularly reliably between the hanging position and the storage position.

(12) In this context it has been shown to be further advantageous if at least one guide element is provided on the deflector element, by means of which guide element the flexible storage elements or storage straps are retained spaced apart from one another in the storage position.

(13) Further advantages and details of the invention result from the following description of preferred exemplary embodiments and on the basis of the drawings. The features and combinations of features mentioned above in the description and the features and combinations of features mentioned below in the description of the figures and/or shown in the figures alone can be used not only in the respective given combination but also in other combinations or in isolation, without departing from the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows a schematic and perspective rear view of a notchback passenger car having a stowage space which can be closed by a stowage space lid shown here in its open position, on the inside of which is arranged a wardrobe hanging means on which items of clothing can be hung in a hanging position when the stowage space lid is open;

(2) FIG. 2 shows a schematic side view of the passenger car in the area of the stowage space with the wardrobe, the hanging means of which as per FIG. 1 respective items of clothing are arranged in a hanging position;

(3) FIG. 3 shows a schematic side view of the passenger car as per FIG. 2, wherein a storage device can be seen within the stowage space with respective stowage straps, the length of which is shortened by means of a retraction device;

(4) FIG. 4 shows a further side view of the passenger car analogous to FIGS. 2 and 3, wherein in this case, when the stowage space lid is closed, the items of clothing can be displaced by means of the storage device from the hanging position to a storage position at least substantially on the underside of the inside of the stowage space lid and/or of an upper side of the stowage space;

(5) FIG. 5 shows a further side view of the passenger car analogous to FIGS. 2 to 4, wherein in this case the stowage space lid is arranged in its closed position in which the items of clothing are

accommodated in their storage position at least substantially on the underside of the inside of the stowage space lid and/or of the upper side of the stowage space; and

(6) FIG. 6 shows a schematic and perspective rear view of the passenger car analogous to FIG. 1, wherein in this case, analogous to FIG. 5, the stowage space lid is illustrated in its closed position in which the items of clothing are arranged by means of the storage device in their storage position at least substantially on the underside of the inside of the stowage space lid and/or of the upper side of the stowage space.

DETAILED DESCRIPTION OF THE DRAWINGS

(7) FIG. 1 illustrates a schematic and perspective rear view of a notchback passenger car. This passenger car comprises a stowage space **10** at the rear which can be closed by a stowage space lid or rear lid **12** that belongs to a wardrobe and is shown here in its open position.

(8) When viewed together with FIG. 2, which shows the saloon in a schematic side view analogous to FIG. 1 with the stowage space lid **12** open, it can be seen that the stowage space lid **12** comprises an area **16** which, in the closed position, adjoins a rear window **14** and overruns approximately horizontally and an area **18** which is angled with respect to the area **16** and extends level with a rear wall. On an inner side of the stowage space lid **12** in the area **18**, a wardrobe provides a hanging means **2**, for example in the form of an eyelet, a bracket, a hook or the like, on which at least one item of clothing **24** can be hung or arranged in a hanging position when the stowage space lid **12** is open. This can be done, for example, with the aid of clothes hangers or the like. As can be seen in FIGS. 1 and 2, the lower end of the respective items of clothing **24** hangs down into the area of the open stowage space **10**.

(9) In the present case, the stowage space lid **12** is pivotably mounted on the body by respective brackets **26** in order to effect the pivoting movement between its open position shown here and its closed position.

(10) In addition to the stowage space lid **12** with the hanging means **22**, the wardrobe comprises a storage device **28** which, in the present case, on the one hand has two flexible storage elements in the form of respective storage straps **28**, **30** which—as can be seen in FIGS. 2 and 3—extend in the longitudinal direction of the vehicle behind the respective items of clothing **24**.

(11) In the present case, between the respective rear storage straps **28**, **30**, respective transverse storage elements **52** are provided, which are also designed, for example, as flexible straps. Altogether, the storage straps **28**, **30** and the transverse storage elements **52** form a storage element made of textile straps, plastic straps or the like in the manner of a rope ladder or the like.

(12) In the present case, the storage straps **28** and **30** or **32** and **34** are spaced or arranged further apart from one another than the typical width of items of clothing **24**. The items of clothing **24** are actually retained in the present case by the transverse storage elements **52**. Nevertheless, as already explained, it is also conceivable that the respective storage straps **28** and **30** or **32** and **34** are arranged at a smaller respective spacing so that the items of clothing **24** are retained directly by the storage straps **28**, **30** or **32**, **34**.

(13) In addition to the two storage straps **28**, **30**—running behind the items of clothing **24** in the longitudinal direction of the vehicle—and the transverse storage elements **52**, the storage device **20** comprises two further flexible storage elements in the form of respective storage straps **32**, **34** which run in front of the items of clothing **24** in the longitudinal direction of the vehicle, as can also be seen from FIGS. 2 and 3, for example. These two front storage straps **32**, **34** are also connected to each other, for example, by transverse storage elements **52**, as has already been explained in connection with the rear storage straps **28**, **30**. With regard to the materials and designs, reference can also be made to the explanations concerning the rear storage straps **28**, **30**.

(14) Overall, the storage straps **28**, **30** and **32**, **34** thus run in each case on the outside or on the edge of the items of clothing **24**, whereby they come into direct contact in particular with the front or rear transverse storage elements **52**.

(15) Alternatively, both rear storage straps **28**, **30** and/or both front storage straps **32**, **34** can be

spaced apart from each other in relation to the transverse direction of the vehicle in such a way that they can come into direct contact with the items of clothing when they are retracted, as will be explained in more detail below, and not just with the transverse storage elements **52**. In this case, the spacing between both rear storage straps **28, 30** and/or both front storage straps **32, 34** is, for instance, between 20 and 50 centimetres. It is also conceivable that instead of multiple storage straps **28, 30**, only one flexible storage element, for example in the form of a fabric web, is provided.

(16) The respective rear storage straps **28, 30** or front storage straps **32, 34** are each attached to the inside **20** of the stowage space lid **12** with their one end **36** or **38**. By way of example, when the stowage space lid **12** is open, the rear storage straps **28, 30** have their respective ends **36** attached in the area **18** behind the hanging means **22**, whereas the front storage straps **32, 34** have their respective ends **38** attached in the area **18** of the stowage space lid **12** to the front of the items of clothing **24**.

(17) Furthermore, the storage device **27** comprises a retraction device **40** with one or more retracting rollers **42**, which is rotatable about an axis which extends at least substantially in the transverse direction of the vehicle and horizontally in the area of an upper side **44** of the stowage space **10** and to the front in the stowage space lid **12**. The respective storage straps **28, 30** or **32, 34** are rolled onto the one or more retracting rollers **42** with ends directed away from their respective ends **36** or **38**.

(18) In addition, the retraction device **40** comprises a deflecting roller **46** or, where necessary, a plurality of deflecting rollers **46** around which, in particular, the front storage straps **32, 34** can be deflected around a corner area **48** on the upper side **44** of the stowage space **12**, which area is adjacent to a front edge of the stowage space lid **12**.

(19) As can now initially be seen when viewing FIGS. **1** and **2** together, the flexible storage straps **28, 30** at the rear of the items of clothing **24** and the storage straps to the front of the items of clothing **24** are unwound with an appropriate amount of slack when the stowage space lid **12** is open, so that respective items of clothing **24** can be conveniently hung on or removed from the hanging means **22**.

(20) When viewing FIGS. **2** and **3** together, it can be seen that in an early phase of a closing movement of the stowage space lid **12**, the respective storage straps **28, 30, 32, 34** and the transverse storage elements **52** are retracted around the retracting roller **42** by means of the retraction device **40**. This can take place, for example, in reaction to the closing movement of the stowage space lid **12** starting. Other mechanisms are also conceivable here, for example spring-loaded pull-out mechanisms. As can be seen in FIG. **3**, the retraction of the respective storage straps **28** to **34** brings in particular the rear storage straps **28, 30** and/or their transverse storage elements **52** into contact with the items of clothing **24** or the rearmost item of clothing **24**.

(21) In the further course of the closing movement of the stowage space lid **12**, which can be seen in FIGS. **4** and **5** in side views analogous to FIGS. **2** and **3**, a further retraction of both the rear storage straps **28, 30** and the front storage straps **34, 36** takes place when closing the stowage space lid **12**. In particular, the rear storage straps **28, 30** and their and the associated transverse storage elements **52** ensure that the items of clothing **24** are moved or displaced from hanging position when the stowage space lid **12** is open as shown in FIGS. **1** to **3** to an inclined intermediate position (FIG. **4**) or to a storage position which they assume when the stowage space lid **12** is at least approximately closed as per FIG. **5**. In this storage position, the items of clothing **24** are located substantially on the underside of the inside of the stowage space lid **12**, to be more precise underneath the inside **20** of the area **16** of the stowage space lid **12**, and below the upper side **44** of the stowage space **10** in the case of relatively long items of clothing **24**. In this case, the items of clothing rest from above on the rear and now lower storage straps **28, 30**. In this case, the items of clothing rest from above on the rear, now lower storage straps **28, 30** and/or their associated transverse storage elements **52**.

(22) The essential function of the front or, in the storage position, upper storage straps **32, 34** and their transverse storage elements **52** is to deflect the items of clothing **24** around the corner area **48** at the rear end of the upper side **44** of the stowage space **10**, which is adjacent to the front edge of the stowage space lid **12**, and thus to ensure that the items of clothing **24** are moved with their respective lower ends collision-free and properly underneath the upper side **44** of the stowage space **10**. In addition, the front, and in the storage position, upper storage straps **32, 34** also serve to clamp the items of clothing **24** between them and the rear or, in the storage position, lower storage straps **28, 30**.

(23) Viewing FIGS. **4** and **5** together also shows that at least one or more guide elements **50** are provided in the deflecting roller **46**, by means of which the flexible storage straps **28, 30** are retained spaced apart from one another relative to the storage straps **32, 34** in the storage position. In the area of the other ends **36, 38**, the storage straps **28, 30** or **32, 34** are spaced apart by their corresponding attachment points.

(24) In FIG. **6**, a schematic and perspective rear view analogous to FIG. **1** shows the saloon again with the stowage space lid **12** closed. In the present case, the respective transverse storage elements **52** can be seen between the respective rear and lower storage straps **28, 39** and/or between the front and upper storage straps **32, 34**. It can be seen once again that the respective storage straps **28, 30** and **32, 34** are spaced or arranged further apart from one another than the typical width of items of clothing **24**. In the present case, the items of clothing **24** are also retained by the transverse storage elements **52**. Nevertheless, as already explained, it is also conceivable that the respective storage straps **28** and **30** or **32** and **34** are arranged at a narrower respective spacing so that the items of clothing **24** are retained directly by the storage straps **28, 39** or **32, 34**.

(25) As already explained, a wider storage web or similar storage element can be used instead of the storage straps **28, 30** or **32, 34**. Overall, it can be seen that the items of clothing **24** are arranged in their storage position in an upper area of the stowage space **10** on the underside of the inside **20** of the area **16** of the stowage space flap **12** or underneath the upper side **44** of the stowage space **10** (in the area below a parcel shelf). The described wardrobe can thus accommodate the items of clothing **24** above a floor of the stowage space **10**, whereby the items of clothing **12** can be isolated from the other luggage, can be kept clean and crease-free and, in addition, it is possible to insert and remove the clothing easily. In addition, an efficient use of space of the luggage compartment or stowage space **10** is thus provided.

(26) This wardrobe is not limited exclusively to notchback vehicles. Such a wardrobe is conceivable in estate cars, coupé vehicles or convertibles.

Claims

1. A wardrobe of a stowage space (**10**) of a passenger car, comprising: a stowage space lid (**12**), wherein a hanger (**22**) is disposed on an inside (**20**) of the stowage space lid (**12**) and wherein an item of clothing (**24**) is hangable on the hanger (**22**) in a hanging position when the stowage space lid (**12**) is open; and a storage device (**27**), wherein the item of clothing (**24**) is displaceable by the storage device (**27**), when the stowage space lid (**12**) is closed, from the hanging position to a storage position on an underside of the inside (**20**) of the stowage space lid (**12**) and/or below an upper side (**44**) of the stowage space (**10**); wherein the storage device (**27**) has a first flexible storage element (**28, 30**) and wherein a length of the first flexible storage element (**28, 30**) is changeable by a retraction device (**40**); wherein the first flexible storage element (**28, 30**) is comprised of a first set of two storage straps (**28, 30**).

2. The wardrobe according to claim 1, wherein a second flexible storage element (**32, 34**) is disposed on a side of the item of clothing (**24**) opposite the first flexible storage element (**28, 30**), wherein a length of the second flexible storage element (**32, 34**) is changeable by the retraction device (**40**), and wherein the item of clothing (**24**) is arrangeable in the storage position between

the first flexible storage element (28, 30) and the second flexible storage element (32, 34).

3. The wardrobe according to claim 2, wherein the second flexible storage element (32, 34) is comprised of a second set of two storage straps (32, 34).

4. The wardrobe according to claim 1, wherein a transverse storage element (52) extends between the two storage straps (28, 30) of the first set of two storage straps (28, 30).

5. The wardrobe according to claim 4, wherein the transverse storage element (52) is flexible.

6. The wardrobe according to claim 3, wherein the storage device (27) has a deflector element (46) for the second flexible storage element (32, 34).

7. The wardrobe according to claim 6, wherein a guide element (50) is disposed on the deflector element (46) and wherein the two storage straps (32, 34) of the second set of two storage straps (32, 34) are retained spaced apart from one another in the storage position by the guide element (50).

8. A wardrobe of a stowage space (10) of a passenger car, comprising: a stowage space lid (12), wherein a hanger (22) is disposed on an inside (20) of the stowage space lid (12) and wherein an item of clothing (24) is hangable on the hanger (22) in a hanging position when the stowage space lid (12) is open; and a storage device (27), wherein the item of clothing (24) is displaceable by the storage device (27), when the stowage space lid (12) is closed, from the hanging position to a storage position on an underside of the inside (20) of the stowage space lid (12) and/or below an upper side (44) of the stowage space (10); wherein the storage device (27) has a first flexible storage element (28, 30) and wherein a length of the first flexible storage element (28, 30) is changeable by a retraction device (40); wherein a second flexible storage element (32, 34) is disposed on a side of the item of clothing (24) opposite the first flexible storage element (28, 30), wherein a length of the second flexible storage element (32, 34) is changeable by the retraction device (40), and wherein the item of clothing (24) is arrangeable in the storage position between the first flexible storage element (28, 30) and the second flexible storage element (32, 34).

9. The wardrobe according to claim 8, wherein the first flexible storage element (28, 30) is comprised of a first set of two storage straps (28, 30).

10. The wardrobe according to claim 8, wherein the second flexible storage element (32, 34) is comprised of a second set of two storage straps (32, 34).

11. The wardrobe according to claim 9, wherein a transverse storage element (52) extends between the two storage straps (28, 30) of the first set of two storage straps (28, 30).

12. The wardrobe according to claim 11, wherein the transverse storage element (52) is flexible.

13. The wardrobe according to claim 10, wherein the storage device (27) has a deflector element (46) for the second flexible storage element (32, 34).

14. The wardrobe according to claim 13, wherein a guide element (50) is disposed on the deflector element (46) and wherein the two storage straps (32, 34) of the second set of two storage straps (32, 34) are retained spaced apart from one another in the storage position by the guide element (50).
